Example 7.10

If the chance of running a bus service according to schedule is 0.8, calculate the probability on a day schedule with 10 services : (i) exactly one is late (ii) atleast one is late

Solution :

Probability of bus running late is denoted as $p = 1-0.8 = 0.2$

Probability of bus running according to the schedule is $q = 0.8$

Also given that $n = 10$

The binomial distribution is

$$p(x) = 10C_x (0.2)^x (0.8)^{10-x}$$

(i) probability that exactly one is late

$$P(x=1) = 10C_1\, pq^9$$

$$= 10C_1 (0.2)(0.8)^9$$

(ii) probability that at least one is late

= 1 – probability that none is late

$= 1 - p(x=0)$

$= 1 - (0.8)^{10}$

Example 7.11

The sum and product of the mean and variance of a binomial distribution are 24 and 128. Find the distribution.

Solution:

For binomial distribution the mean is np and varaiance is npq

Given values are

$np + npq = 24 \quad np(1 + q) = 24$ – (1)

Other term

$np \times npq = 128 \quad n^2p^2q = 128$ – (2)

From (1) we get $np = 24/(1+q)$ which implies $n^2p^2 = (24/(1+q))^2$

Substitute this value in equation (2) we get

$$\left(\frac{24}{1+q}\right)^2 q = 128 \text{ which implies } 9q = 2(1+2q+q^2)$$

$(2q - 1)(q - 2) = 0$

Where $q = \frac{1}{2}$ and $p = \frac{1}{2}$

Substitute in (1) we get $n = 32$

Hence the binomial distribution

$$32C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{32-x}$$

Example 7.12

Suppose A and B are two equally strong table tennis players. Which of the following two events is more probable:

(a) A beats B exactly in 3 games out of 4 or

(b) A beats B exactly in 5 games out of 8 ?

Solution :

Here $p = q = 1/2$

(a) probability of A beating B in exactly 3 games out of 4

$$\binom{4}{3}\left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{4-3}$$

= ¼ = 25%

(b) probability of A beating B in exactly 5 games out of 8

$$\binom{8}{5}\left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{8-5}$$

$$= \frac{7}{32} = 21.875\%$$

Clearly, the first event is more probable.

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